SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question 1 – Rain and Wet Ground

Knowledge Base:

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal: Prove that the ground is wet using the Resolution Algorithm.

Step 1: Represent the statements in propositional logic.

R = It is raining.

W = The ground is wet.

Statement 1: $R \rightarrow W$

Statement 2: R

Step 2: Convert the statements into Conjunctive Normal Form (CNF).

$R \rightarrow W$ is equivalent to $\neg R \lor W$.

R is already in CNF.

CNF clauses: C1 = $(\neg R \lor W)$, C2 = (R)

Step 3: Negate the goal.

Goal: W

Negated goal: $\neg W$

Add C3 = $(\neg W)$ to the set of clauses.

Step 4: Apply the Resolution Algorithm.

Resolve C1 = $(\neg R \lor W)$ with C2 = (R). The complementary literals $\neg R$ and R cancel, giving C4 = (W).

Resolve C4 = (W) with C3 = $(\neg W)$. The complementary literals W and $\neg W$ cancel, producing the empty clause □.

Conclusion: Since the empty clause □ is derived, the negation of the goal leads to a contradiction. Therefore, the goal is proved: the ground is wet.

Question 2 – Student Assignment Submission

Knowledge Base:

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal: Prove that Rahul receives internal marks using the Resolution Algorithm.

Step 1: Propositional representation.

S = Rahul submitted the assignment.

M = Rahul receives internal marks.

Statement 1: $S \rightarrow M$

Statement 2: S

Step 2: Convert to CNF.

$S \rightarrow M$ becomes $\neg S \lor M$.

S remains S.

C1 = $(\neg S \lor M)$, C2 = (S)

Step 3: Negate the goal.

Goal: M

Negated goal: $\neg M$

C3 = $(\neg M)$

Step 4: Resolution.

Resolve C1 = $(\neg S \lor M)$ with C2 = (S). Resolving S and $\neg S$ gives C4 = (M).

Resolve C4 = (M) with C3 = $(\neg M)$. Resolving M and $\neg M$ produces the empty clause □.

Conclusion: The empty clause is obtained, so the negated goal is inconsistent with the knowledge base. Therefore, Rahul receives internal marks.

Question 3 – Library Membership

Knowledge Base:

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal: Prove that Priya can borrow books using the Resolution Algorithm.

Step 1: Propositional representation.

L = Priya is a library member.

B = Priya can borrow books.

Statement 1: $L \rightarrow B$

Statement 2: L

Step 2: Convert to CNF.

$L \rightarrow B$ becomes $\neg L \lor B$.

L is already a CNF clause.

C1 = $(\neg L \lor B)$, C2 = (L)

Step 3: Negate the goal.

Goal: B

Negated goal: $\neg B$

C3 = $(\neg B)$

Step 4: Resolution.

Resolve C1 = $(\neg L \lor B)$ with C2 = (L). The complementary literals L and $\neg L$ resolve to give C4 = (B).

Resolve C4 = (B) with C3 = $(\neg B)$. The result is the empty clause □.

Conclusion: Since the empty clause is derived, the goal is logically proved. Therefore, Priya can borrow books.

Question 4 – Placement Eligibility

Knowledge Base:

1. If the student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal: Prove that Arun is eligible for placement using the Resolution Algorithm.

Step 1: Propositional representation.

A = Arun cleared the aptitude test.

E = Arun is eligible for placement.

Statement 1: $A \rightarrow E$

Statement 2: A

Step 2: Convert to CNF.

$A \rightarrow E$ becomes $\neg A \lor E$.

A is already in CNF.

C1 = $(\neg A \lor E)$, C2 = (A)

Step 3: Negate the goal.

Goal: E

Negated goal: $\neg E$

C3 = $(\neg E)$

Step 4: Resolution.

Resolve C1 = $(\neg A \lor E)$ with C2 = (A). The complementary literals $\neg A$ and A cancel, producing C4 = (E).

Resolve C4 = (E) with C3 = (¬E). The complementary literals E and ¬E produce the empty clause □.

Conclusion: The empty clause is obtained, proving that the negation of the goal is impossible. Therefore, Arun is eligible for placement.

Question 5 – Access Control System

Knowledge Base:

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal: Prove that the user is granted access using the Resolution Algorithm.

Step 1: Represent the knowledge base in propositional logic.

P = The user entered the correct password.

A = The user is authenticated.

G = The user is granted access.

Statement 1: $P \rightarrow A$

Statement 2: $A \rightarrow G$

Statement 3: P

Step 2: Convert all statements into CNF.

$P \rightarrow A$ becomes $\neg P \lor A$.

$A \rightarrow G$ becomes $\neg A \lor G$.

P is already a CNF clause.

C1 = $(\neg P \lor A)$

C2 = $(\neg A \lor G)$

C3 = (P)

Step 3: Negate the goal.

Goal: G – The user is granted access.

Negated goal: $\neg G$

C4 = $(\neg G)$

Step 4: Apply the Resolution Algorithm.

Resolve C1 = $(\neg P \lor A)$ with C3 = (P). The complementary literals ¬P and P cancel, giving C5 = (A).

Resolve C2 = $(\neg A \lor G)$ with C5 = (A). The complementary literals ¬A and A cancel, giving C6 = (G).

Resolve C6 = (G) with C4 = (¬G). The complementary literals G and ¬G cancel, producing the empty clause □.

Resolution proof summary:

$(\neg P \lor A), P \Rightarrow A$
$(\neg A \lor G), A \Rightarrow G$
$G, \neg G \Rightarrow \square$

Final Conclusion: The empty clause □ has been derived. Therefore, the negation of the goal is contradictory to the knowledge base, proving that the user is granted access. The logical chain is: correct password → authenticated → access.

Final Note on the Resolution Method

In all five problems, the resolution proof follows the same basic procedure: represent the knowledge base in propositional logic, convert implications into CNF clauses, negate the required goal, add the negated goal to the clause set, and repeatedly apply the resolution rule. If the empty clause □ is derived, the negated goal is contradictory and the original goal is proved. If no empty clause can be derived, the goal cannot be established from

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation